

MARY CLAYTON SOTO

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EDUCATION

Duke University, *Pratt School of Engineering*

Aug 2024 – B.S. 2028

- **Majors:** Biomedical Engineering & Mathematics; **Minor:** Visual Arts
- **GPA:** 3.92/4.00
- **Activities & Societies:** Pratt Tour Guide, Club Soccer, Theta Tau Engineering Fraternity, Phi Alpha Delta, Tri Delta
- **Relevant Coursework:** Quantitative Physiology with Biostatistical Applications, Engineering Design and Collaboration, Mechanics of Solids, Materials Science, Signals and Systems, Modeling Cellular and Molecular Systems

Maclay School, *Salutatorian*

May 2024

ENGINEERING EXPERIENCE

Duke Surgical Education & Assessment Lab, *Research Assistant*

Mar 2025 – Present

- Design and iterate hydrogel models across 3 organ types from MRI-derived DICOM scans using 3D Slicer segmentation and Fusion 360 molds, tuning materials to replicate human tissue mechanics for surgical simulation.
- Develop and validate a computer-vision system that scores surgical knot-tying from video, evaluated on 1,000+ recordings.

FSU Microparticle Lab, *Research Assistant*

May – Aug 2025

- Developed a MATLAB pipeline to segment and classify fluorescence cell images, analyzing ~20,000 cells and generating a standardized dataset for peer-reviewed publication.
- Automated cell quantification through optimized convolution-based segmentation and thresholding, improving classification accuracy across heterogeneous fluorescence channels and saving hundreds of hours of manual counting.

Tallahassee Research Institute, *Research Coordinator*

May – Aug 2026

- Managed clinical research workflows across 10+ cardiovascular studies, coordinating protocols and documentation.
- Compiled and analyzed patient data into structured reports to support ongoing device trials and investigator decision-making.

ACADEMIC PROJECTS

Rollator Design Project

Aug 2024 – May 2025

- Prototyped a wheeled-cage oxygen canister mounting system for Duke Hospital rollators, enabling post-surgery patients to walk independently and freeing a second staff member previously required to roll the oxygen.
- Validated against 7 design criteria, reducing tipping angle by 6° and earning a 4.3/5 usability rating from 30 evaluators.

Biomedical Image Segmentation Pipeline

Jan – May 2026

- Built and compared three Python pipelines for biological cell image segmentation, applying Fourier transform analysis, convolution-based edge detection, and thresholding, and found convolution alone captured most cell boundaries while frequency enhancement improved low-contrast detail.

Computational Modeling of Engineered Probiotic Therapeutics

Jan – May 2026

- Built a multi-scale computational model of an engineered yeast system designed to degrade PET microplastics in the GI tract, integrating gene regulation, Michaelis-Menten enzyme kinetics, and gut transport equations to evaluate system-level degradation efficiency.

Yeast Genomic Library Project

Aug – Dec 2025

- Executed a full recombinant DNA workflow, from yeast genomic DNA isolation and plasmid cloning to bacterial transformation, validating constructs by PCR, gel electrophoresis, Sanger sequencing, and bioinformatic analysis.

LEADERSHIP & INVOLVEMENT

EGR105: Coding for Engineers, *Teaching Assistant*

Aug 2025 – May 2026

- Mentored 50+ first-year engineering students in Python algorithm design, reinforcing debugging workflows and Git version control.
- Taught NumPy and Matplotlib for data analysis and visualization, emphasizing efficient data handling and interpretation.

Theta Tau, *Vice President of Professional Development*

Aug 2025 – Present

- Execute events to highlight engineering research and innovation, expanding 100+ members' professional opportunities and skills.
- Created a centralized member resource database for academic planning, networking, and career development.

PUBLICATIONS & PRESENTATIONS

- Fukuda, M.; Soto, M. C.; et al. "Engineered Microparticles Suggest Proteolysis as a Prerequisite for Chromatin Clearance in Macrophage Phagosomes." *ACS Appl. Mater. Interfaces*, 2026, 18, 16137–16147.
- Gao, S.; Yamamoto, K. K.; Soto, M. C.; et al. "Dissolvable Core Molding (DCM): A Gallbladder Model Toward Fluid-Bearing Organ Simulators." *ACS Surgical Simulation Summit*, 2027 (accepted).

TECHNICAL SKILLS

- **Languages & Software:** Python, MATLAB, R, Git, Fusion 360, SolidWorks, 3D Slicer, DICOM segmentation, Meshmixer, C++
- **Fabrication:** 3D printing (FDM, resin), laser & waterjet cutting, silicone molding & casting
- **Analysis & Modeling:** image processing, correlation studies, data synthesis, ANOVA, regression, computational modeling